

When the above command is executed, it generates the following output:

```
data.iso
sent 1.07G bytes received 35 bytes 126.35M bytes/sec # This
indicated entire file is copied
total size is 1.07G speedup is 1.00
```

The above output indicates that the entire file was copied, as this copying was done for the first time. Now let us try to copy the same file:

```
[root]# rsync -vh --no-W data.iso /opt/backup/
```

When the above command is executed, it generates the following output:

```
data.iso
sent 131.16K bytes received 229.42K bytes 24.87K bytes/sec
# This indicated only 131.16K bytes are copied
total size is 1.07G speedup is 2,977.76
```

If you compare the above output with the previous output, then you will observe that this time only 131.16K bytes of data was transferred.

Now, let us demystify the command. In the above command:

- The `-v` flag is used to show verbose output.
- The `-h` flag is used to show output in human readable format, e.g., in KB, MB, GB, and so on.
- The `--no-W` flag is used to enable the delta transfer algorithm.

b) Enable compression: rsync supports various compression configurations. We can enable the default compression level using the `-z` option. It will compress file data during the transfer. We can use it as follows:

```
[root]# rsync -zvh --no-W data.iso /opt/backup/
```

We can also set the compression level explicitly using the `--compress-level` option. This option accepts compression level as a digit from 0-9, where 1 signifies the least compressed and 9 the most compressed. We can use this option as follows:

```
[root]# rsync -zvh --no-W --compress-level=5 data.iso /opt/
backup/
```

In addition, we can disable compression for certain file types. We can use the `--skip-compress` option for this.

This option accepts the suffix list for which compression is disabled. We can use this option as follows:

```
[root]# rsync -zvh --no-W --skip-compress=7z/bz2/gz data.iso
/opt/backup/
```

c) Show progress: The rsync operation will take some time depending upon the file size. In such cases, showing progress information is very helpful. It supports the `--progress` option, which shows progress during the file transfer. We can use this option as follows:

```
[root]# rsync -rvh data/data-1.iso /opt/backup/ --progress
```

When the above command is executed, it generates the following output:

```
sending incremental file list
data-1.iso
1.07G 100% 120.27MB/s 0:00:08 {xfr#1, to-chk=0/1}

sent 1.07G bytes received 35 bytes 126.35M bytes/sec
total size is 1.07G speedup is 1.00
```

d) Recursive mode: So far we have synchronised a single file. However, rsync supports the recursive mode, which allows synchronisation of all files from the directory. We can use the `-r` option for this as follows:

```
[root]# rsync -rvh data/ /opt/backup/
```

When the above command is executed, it generates the following output:

```
sending incremental file list
data-1.iso
data-2.iso
data-3.iso

sent 3.22G bytes received 73 bytes 121.59M bytes/sec
total size is 3.22G speedup is 1.00
```

e) Preserve timestamp: Let us compare timestamps of the source and the destination files:

```
[root]# ls -lrth
total 1.0G
-rw-r--r--. 1 root root 1.0G Dec 6 17:46 data.iso
```

```
[root]# ls -lrth /opt/backup/data.iso
-rw-r--r--. 1 root root 1.0G Dec 6 18:21 /opt/backup/
data.iso
```